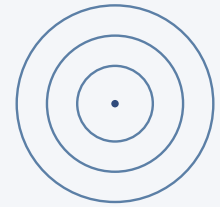


BLADE3000A

QUANTUM SOFTWARE DEVELOPER



I design and build software at the intersection of classical engineering and quantum computing — translating concepts such as superposition, entanglement, and quantum measurement into working, testable code.

SELECTED WORK

MULTICLASS QML

Beyond Built-in Multiclass — Binary VQC Decomposition on Iris

A quantum classification pipeline built on Qiskit's Variational Quantum Classifier, redesigned to make more confident decisions on ambiguous cases. Three binary quantum classifiers vote in parallel, and a second layer resolves any disagreement, with classical baselines kept alongside for comparison.

→ github.com/Blade3000A/qiskit-multiclass-vqc-ovr-ovo

QUANTUM CRYPTOGRAPHY

BB84 Quantum Key Distribution

A simulation of the BB84 protocol for secure key exchange, including an eavesdropper positioned on the communication line. The project measures how reliably an intercept-and-resend attack is detected as the eavesdropper's activity increases, validating the protocol's core security guarantee.

→ github.com/Blade3000A/BB84_QuNetSim

QUANTUM NONLOCALITY

The CHSH Game

An implementation of the CHSH game, a cornerstone experiment in the study of quantum nonlocality. The project compares the best possible classical strategy against an entanglement-based quantum strategy, and demonstrates the latter exceeding the classical limit exactly as theory predicts.

→ github.com/Blade3000A/CHSH_QuNetSim

QUANTUM NONLOCALITY

The GHZ Game

An extension of the nonlocal-game framework to three players using a shared GHZ state. Where classical strategies are limited to a 75% win rate, the quantum strategy implemented here reaches a perfect score, offering a clear, reproducible demonstration of quantum correlations with no classical equivalent.

→ github.com/Blade3000A/GHZ_game_QuNetSim

CIRCUIT OPTIMIZATION

Quantum Circuit Optimization with ZX-Calculus

A PennyLane-to-PyZX pipeline that rewrites a quantum circuit as a ZX-diagram, simplifies it using graph-theoretic rewrite rules, and extracts an optimized circuit back out. On the notebook's benchmark circuit, gate count drops from 63 to 37 and T-count from 28 to 8, with output equivalence verified at each step.

→ github.com/Blade3000A/zx-calculus-pennylane